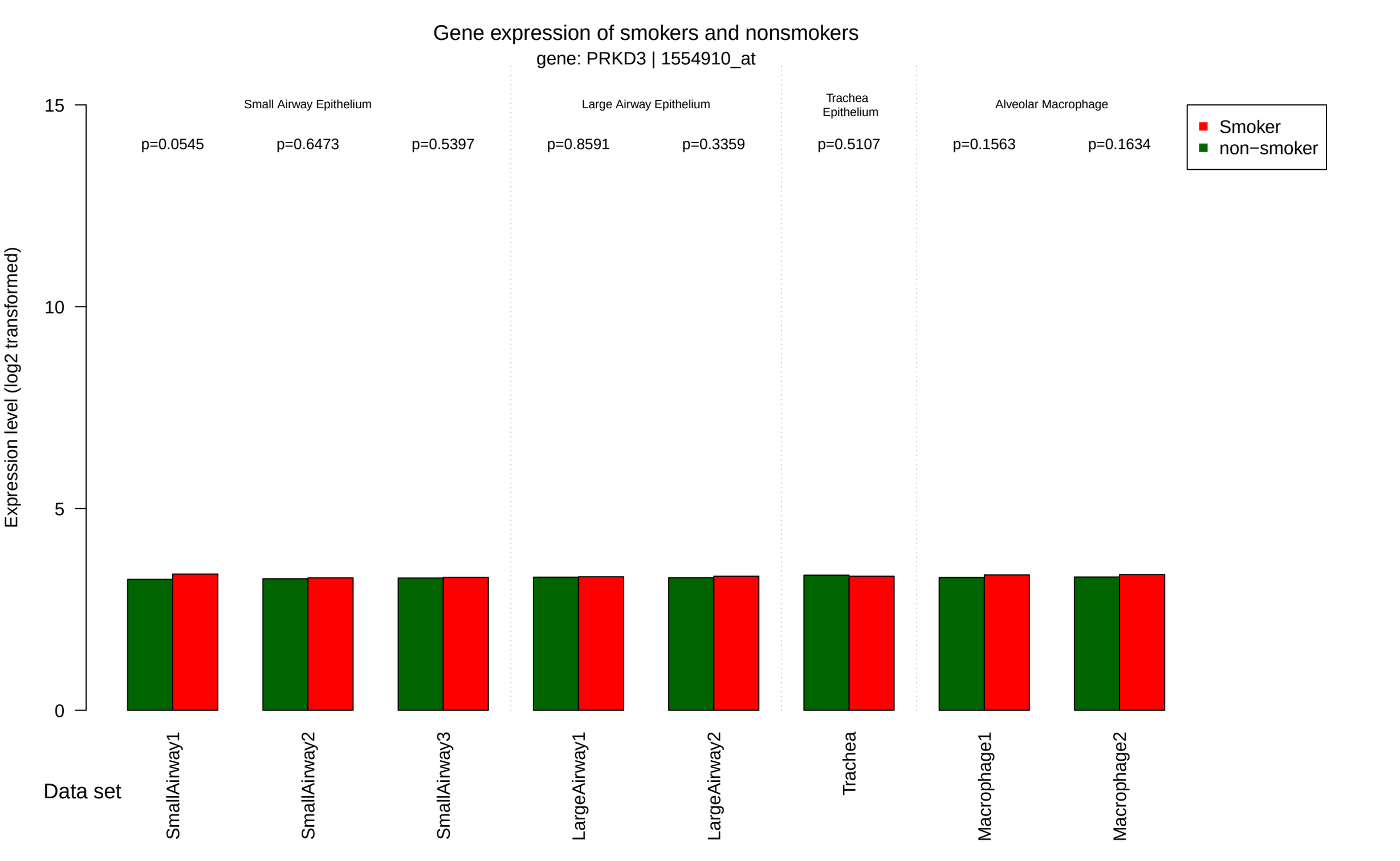
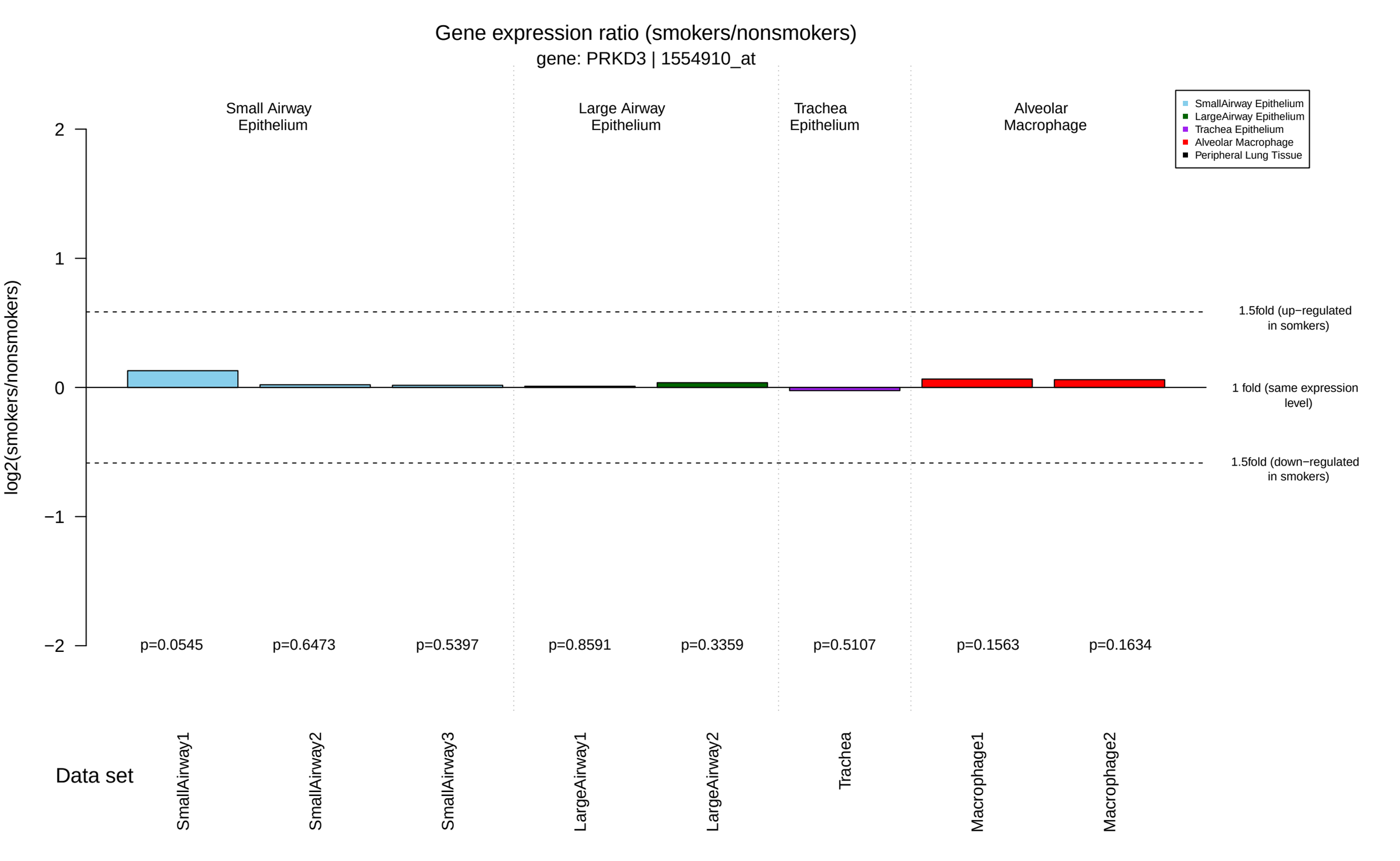
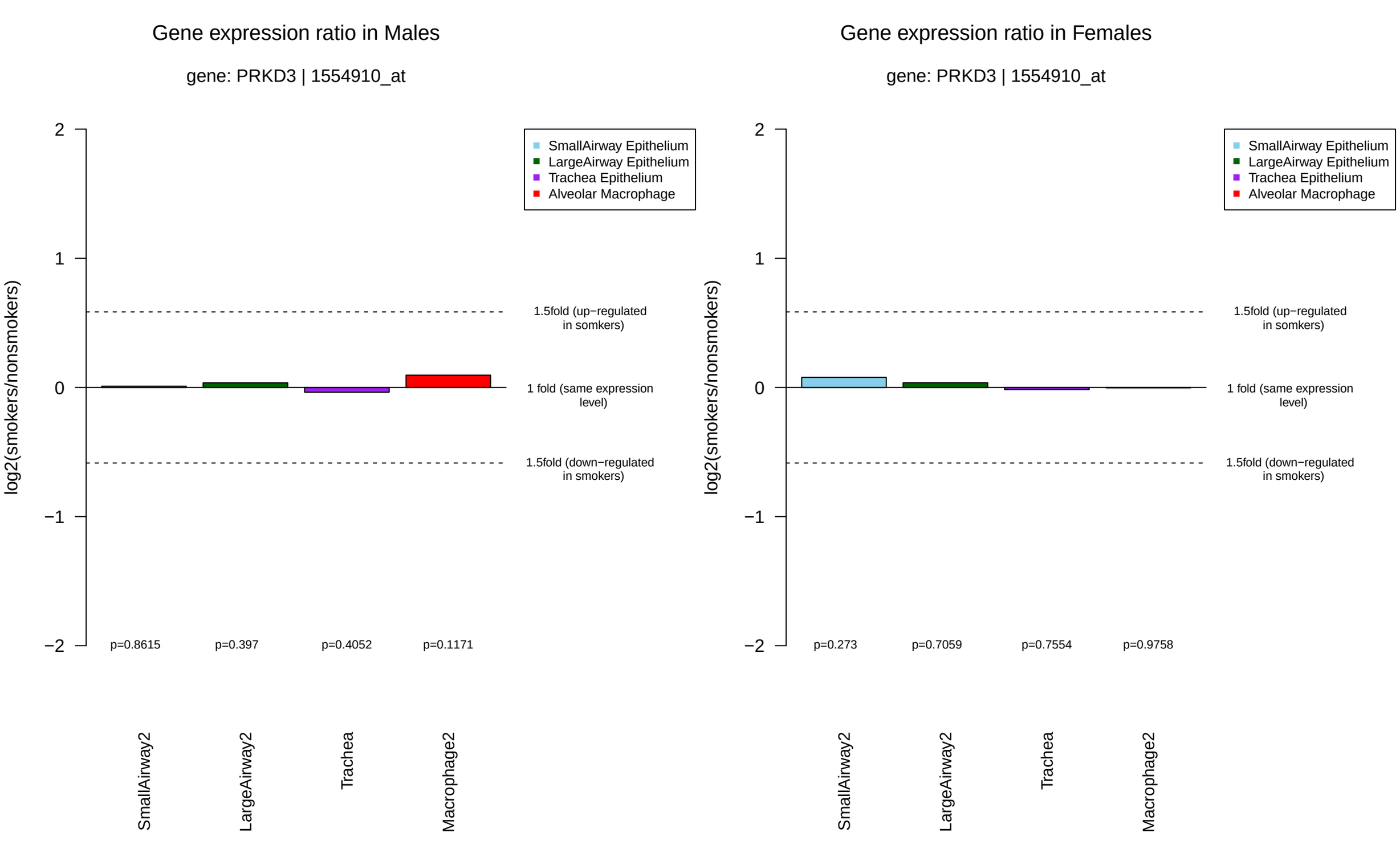
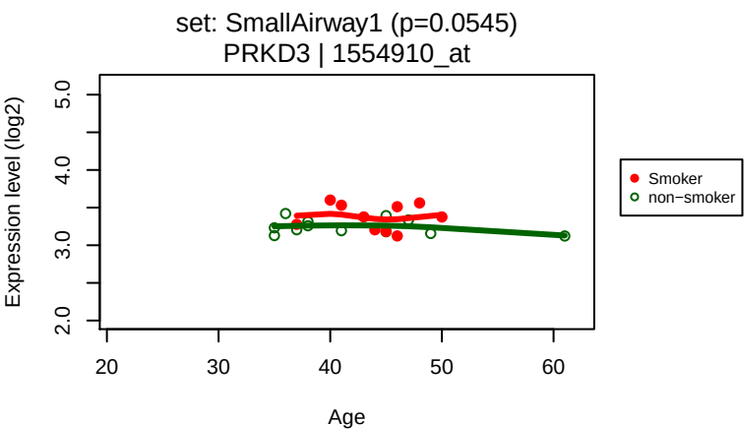
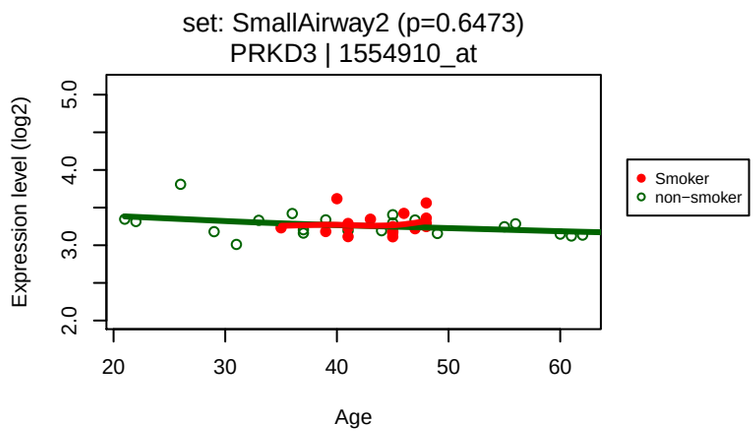
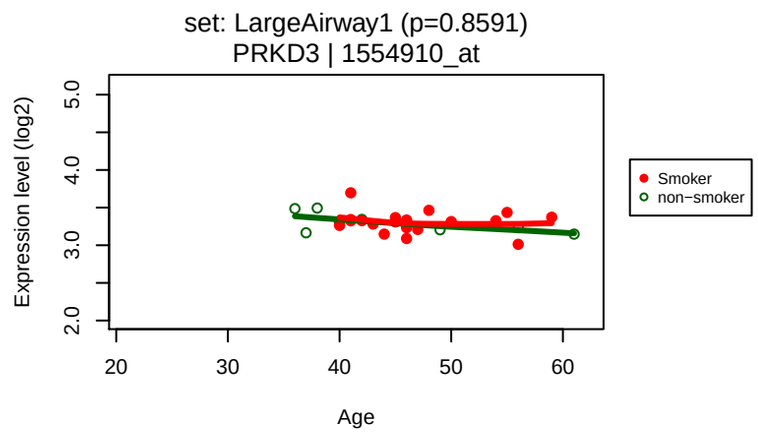

Gene expression of smokers and nonsmokers
gene: PRKD3 | 1554910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0545
p=0.6473
p=0.5397
p=0.8591
p=0.3359
p=0.5107
p=0.1563
p=0.1634
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

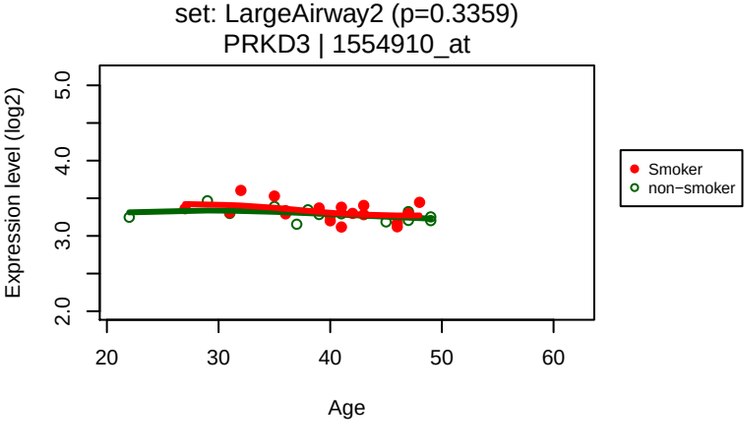

Gene expression ratio (smokers/nonsmokers)
gene: PRKD3 | 1554910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0545
p=0.6473
p=0.5397
p=0.8591
p=0.3359
p=0.5107
p=0.1563
p=0.1634
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

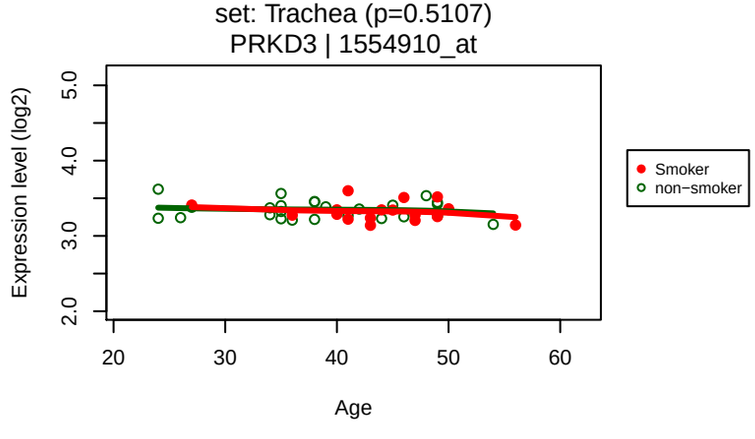

Gene expression ratio in Males
gene: PRKD3 | 1554910_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8615
p=0.397
p=0.4052
p=0.1171
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKD3 | 1554910_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.273
p=0.7059
p=0.7554
p=0.9758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

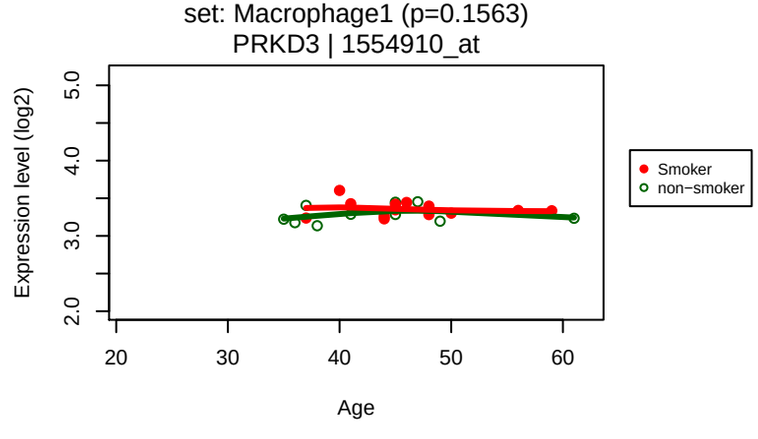

set: SmallAirway1 (p=0.0545)
PRKD3 | 1554910_at
Expression level (log2)
Age
Smoker
non−smoker

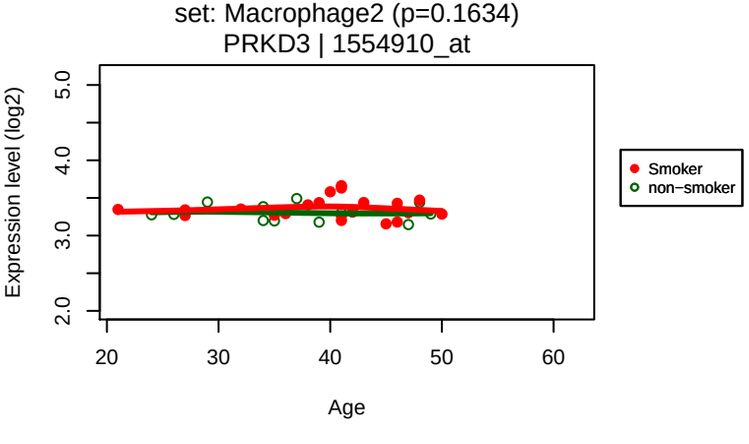

set: SmallAirway2 (p=0.6473)
PRKD3 | 1554910_at
Expression level (log2)
Age
Smoker
non−smoker

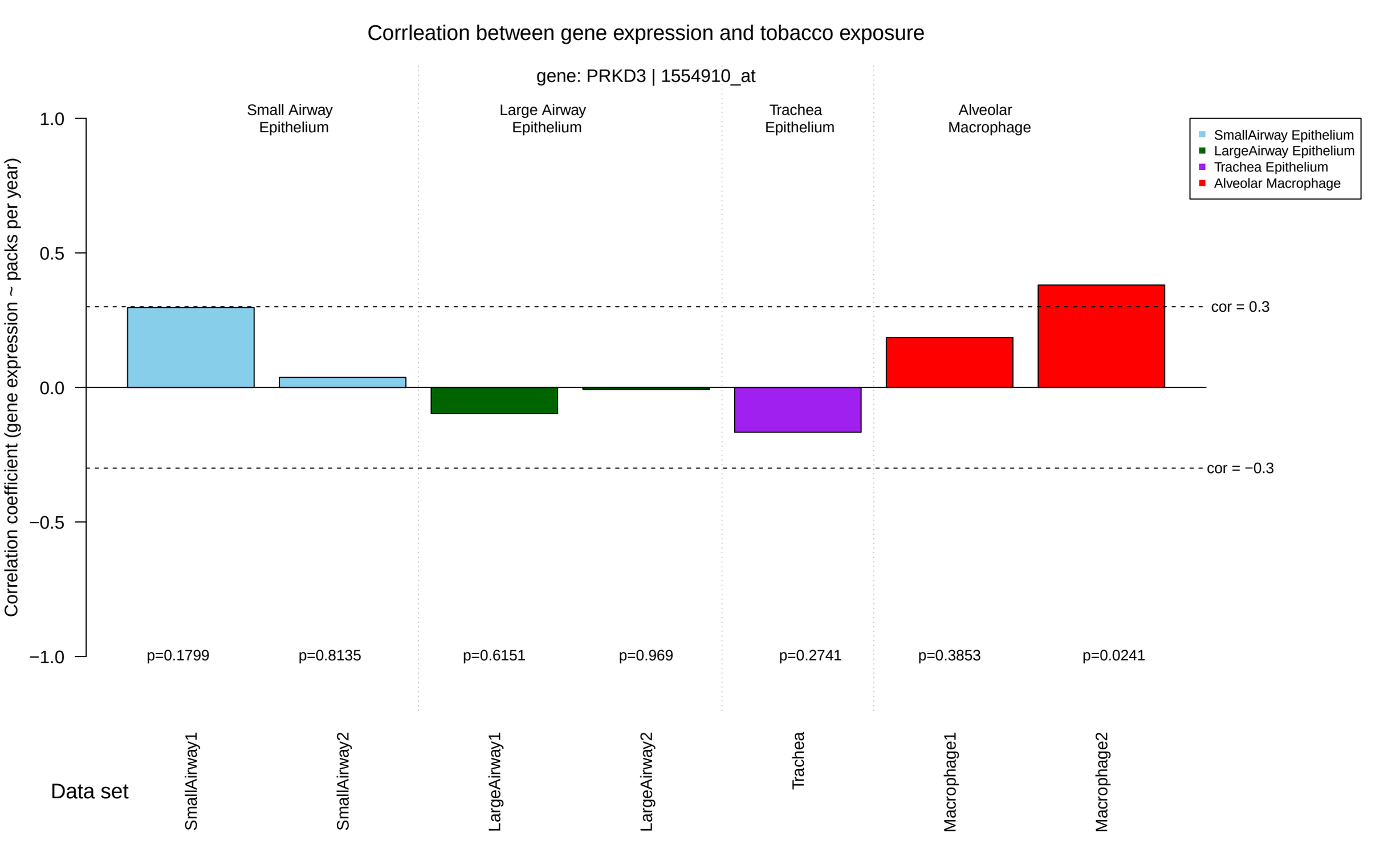

set: LargeAirway1 (p=0.8591)
PRKD3 | 1554910_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3359)
PRKD3 | 1554910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5107)
PRKD3 | 1554910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1563)
PRKD3 | 1554910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1634)
PRKD3 | 1554910_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKD3 | 1554910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1799
p=0.8135
p=0.6151
p=0.969
p=0.2741
p=0.3853
p=0.0241
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2